INCH-POUND

MIL-T-27/175B
20 April 1993
SUPERSEDING
MIL-T-27/175A
12 March 1979

MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE), INDUCTORS, AUDIO FREQUENCY, HIGH Q, TF4R20YY

(B) Inactive for new design after the date of this specification

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet, and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-27

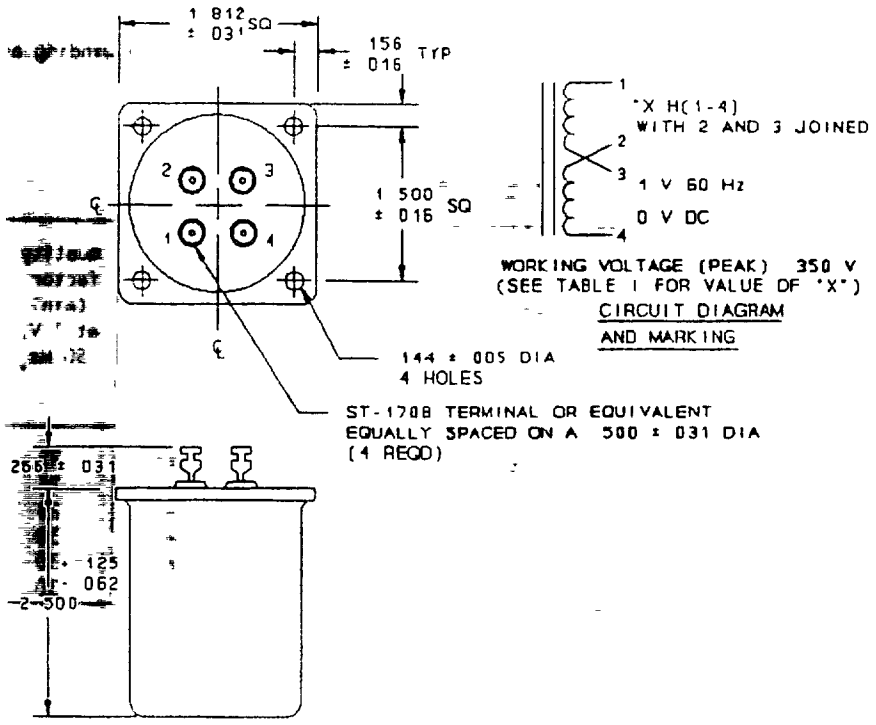


FIGURE 1. Dimensions and configurations

(B) denotes changes

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inches	mm
005	0 13
016	0.41
031	0 79
062	1 57
125	3 18
144	3 66
156	3 96
266	6 76
500	12 70
1 500	38 10
1 812	46 02
2 500	63 50

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Marking shall be on the sides of the case
4 Electrical values shall be as specified in table I, as applicable

FIGURE 1 Dimensions and configurations - Continued

REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the windings)

Electrical ratings See table I

TABLE I Electrical ratings

Dash no	Inductance H ±2 percent		1/ Potential V at 60 Hz	Current ampere	DC resistance ohms ±20% with (2-3) connected	Temperature stability (-55°C to +105°C)	Quality factor (min) at 1 V, 50 Hz,	Working voltage (peak)
	Series connected	parallel connected						
	(1-4)	(1-4)	(1-4)	(1-4)	(1-4)			
01	1 0	0 25	1.0/0 5	0	5 0	±3 percent	25	350
02	10	2 5	1 0/0 5	0	50	±3 percent	25	350
03	20	5	1 0/0 5	0	82	±3 percent	28	350
04	200	50	1 0/0 5	0	820	±3 percent	30	350
05	400	100	1 0/0 5	0	2100	±3 percent	30	350
06	2500	652	1 0/0 5	0	13k	±3 percent	30	350

1/ Values correspond to the winding being connected in series/parallel respectively

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Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Terminals Solder lug type, ST-1708 or equivalent

Terminal height 266 ± 031 inch

Weight 0 454 kilogram, maximum

Operating temperature range -55°C to +105°C

Terminal strength MIL-STD-202, method 211, test condition A, 5 pounds

Shock (specified pulse) MIL-STD-202, method 213, test condition H

Dielectric withstanding voltage (at sea level) 1000 V rms

Electrical characteristics

Inductance.

Series connected With 1 V rms, 60 Hz across (1-4), and 0 ampere dc applied to (1-4) the inductance shall be as specified in table I

Parallel connected With 0 5 V rms, 60 Hz across (1-4), and 0 ampere dc applied to (1-4) the inductance shall be as specified in table I

Q At 1 V, 50 Hz, the Q shall be as specified in table I

(B) Quality assurance provisions

Qualification inspection Not applicable for this specification sheet

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable

Marking location See figure 1

Part or Identifying Number (PIN) M27/175-(dash number from table I)

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0815-16)

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